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POWER SWITCH DEACTIVATION DRIVER

Abstract

One example includes a power supply system. The system includes a power switch configured to activate via a control voltage responsive to a first state of an activation signal to conduct current from a power rail to a switching terminal. The system further includes a power switch deactivation driver configured to control an amplitude of the control voltage responsive to a second state of the activation signal based on a voltage difference between the power rail and the switching terminal to provide for a variable rate of deactivation of the power switch.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 17/708,045 filed Mar. 30, 2022, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This description relates generally to electronic systems, and more particularly to a power switch deactivation driver.

BACKGROUND

[0003] Switching power supplies are widely implemented in circuits to provide efficient power. A typical power supply, such as a DC-DC converter, includes a power stage having a transistor. One such DC-DC converter includes a high-side transistor and a low-side transistor that are coupled between an input terminal $V_{sub.IN}$ and a ground terminal GND, with a switching terminal between the high-side and low-side transistors. The high-side transistor and the low-side transistor can thus be alternately activated to provide current through an inductor coupled to the switching terminal. For example, when the high-side transistor is activated, a current flows from an input terminal on which an input voltage is provided through the high-side transistor to the switching terminal, and when the low-side transistor is activated, a current flows from a ground terminal through the low-side transistor to the switching terminal. Activation and deactivation of the high-side and low-side transistors results in switching loss, which is proportional to an input voltage provided at the input terminal, and inversely proportional to an activation/deactivation speed.

SUMMARY

[0004] One example includes a power supply system. The system includes a power switch configured to activate via a control voltage responsive to a first state of an activation signal to conduct current from a power rail to a switching terminal. The system further includes a power switch deactivation driver configured to control an amplitude of the control voltage responsive to a second state of the activation signal based on a voltage difference between the power rail and the switching terminal to provide for a variable rate of deactivation of the power switch.

[0005] Another example described herein includes a power switch deactivation driver to control a power switch configured to by a first state of an activation signal to conduct current from a power rail to a switching terminal. The power switch deactivation driver includes a sense resistor coupled to the power rail to conduct a sense current having an amplitude based on a voltage difference between the power rail and the switching terminal responsive to a second state of the activation signal opposite the first state. The power switch deactivation driver further includes a shutoff switch configured to control an amplitude of a control voltage configured to control an activation of the power switch. The shutoff switch can be controlled by the amplitude of the sense current to provide for a variable rate of deactivation of the power switch.

[0006] Another example described herein includes an integrated circuit. The circuit includes a power switch comprising an input, a first terminal coupled to a power rail, and a second terminal coupled to a switching terminal. The circuit also includes a power switch deactivation driver. The power switch deactivation driver includes a shutoff switch comprising an input, a first terminal coupled to the input of the power switch, and a second terminal coupled to the switching terminal or the power rail. The power switch deactivation driver also includes a first sense resistor comprising a first terminal coupled to the power rail or the switching terminal and a second terminal and a second sense resistor comprising a first terminal coupled to the input of the shutoff switch. The power switch deactivation driver also includes a first current mirror transistor comprising a first terminal coupled to the first sense resistor and a second terminal coupled to the other of the power rail or the switching terminal. The power switch deactivation driver further

includes a second current mirror transistor comprising a first terminal coupled to the second sense resistor and the input of the shutoff switch, the second current mirror transistor also comprising a second terminal coupled to the other of the power rail or the switching terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is an example block diagram of a power supply system.

[0008] FIG. 2 is an example diagram of a power switch deactivation driver.

[0009] FIG. 3 is an example of a timing diagram.

[0010] FIG. 4 is another example diagram of a power switch deactivation driver.

DETAILED DESCRIPTION

[0011] This description relates generally to electronic systems, and more particularly to a power switch deactivation driver. The power switch deactivation driver can be implemented in any of a variety of power supply circuits. A power supply circuit can include a power switch that is activated via an activation signal to conduct current from one of a power rail and a switching terminal based on a control voltage. The power switch can correspond to one of the high-side and low-side power switches in a DC-DC buck power supply, or can correspond to a low-side power switch in a DC-DC boost power supply. The power switch deactivation driver can be configured to control an amplitude of the control voltage responsive to a second state of the activation signal, with the amplitude of the control voltage being based on a voltage difference between the power rail and the switching terminal to provide for a variable rate of deactivation of the power switch.

[0012] As described herein, the term “activate” with respect to a transistor (including the power switch) refers to a transition of the transistor state from cutoff mode through linear mode to saturation mode. As described herein, the term “deactivate” with respect to a transistor (including the power switch) refers to a transition of the transistor state from saturation mode through linear mode to cutoff mode. As described herein, the term “variable rate of the deactivation” with respect to the power switch refers to a non-linear decrease of the control voltage to deactivate the power switch. As an example, the variable rate of deactivation can include an initially rapid deactivation of the power switch based on a more rapid rate of decrease of the control voltage, followed by a slower deactivation of the power switch based on a slower rate of decrease of the control voltage. The variable rate of deactivation can provide for optimization of efficiency and a reduction of ringing of the current through the power switch during deactivation of the power switch, as described in greater detail herein.

[0013] For example, the power switch deactivation driver can include a first sense transistor that can conduct a first sense current having an amplitude that is based on a difference between the voltages of the power rail and the switching terminal. The power switch deactivation driver also includes a current mirror and a second sense resistor, such that the current mirror generates a second sense current through the second sense resistor, with the second sense current having an amplitude that is proportional to the first sense current. The second sense resistor can be coupled to a control terminal (e.g., gate) of a shutoff transistor. The shutoff transistor can be coupled to the control terminal of the power switch, such that the shutoff transistor can control the deactivation of the power switch. As a result, based on the voltage difference between the power rail and the switching terminal, the shutoff transistor can control the deactivation of the power switch at a variable rate.

[0014] FIG. 1 is an example block diagram of a power supply system **100**. The power supply system **100** can be a DC-DC power supply that generates an output voltage $V_{\text{sub.OUT}}$ based on a high-voltage rail, demonstrated in the example of FIG. 1 as an input voltage $V_{\text{sub.IN}}$, responsive to a set of control signals CTRL. As an example, the power supply system can be arranged as a

buck converter or a boost converter. The power supply system **100** can be included in or as part of an integrated circuit (IC).

[0015] In the example of FIG. 1, the power supply system **100** includes one or more power switches **102**. In the example of the power supply system **100** being arranged as a buck converter, the power switch(es) **102** can include a high-side switch and a low-side switch that are arranged between the input voltage $V_{sub.IN}$ and a low-voltage rail, demonstrated in the example of FIG. 1 as ground, with a switching terminal arranged therebetween. In the example of the power supply system **100** being arranged as a boost converter, the power switch(es) **102** can include a low-side switch that is arranged between the low-voltage rail and the switching terminal. As an example, the power switch(es) **102** can be arranged as N-channel field effect transistors (NFETs). Each of the power switch(es) **102** is activated via an activation signal (e.g., one of the control signals CTRL) to conduct current from one of a power rail and a switching terminal based on a control voltage.

[0016] In the example of FIG. 1, the input voltage $V_{sub.IN}$ is provided to the power supply system **100** via an inductor $L_{sub.P1}$ that represents a parasitic inductance associated with a pin of the associated IC package on which the power supply system **100** is provided (e.g., to receive the input voltage $V_{sub.IN}$ externally). Similarly, the power supply system **100** is coupled to ground via an inductor $L_{sub.P2}$ that represents a parasitic inductance associated with another pin of the associated IC package on which the power supply system **100** is provided (e.g., to couple the IC package to an external ground). Responsive to a rapid deactivation of the power switch(es) **102**, the current that was provided through the respective power switch(es) **102** can ring (e.g., oscillate) based on the inductance of the parasitic inductors $L_{sub.P1}$ and $L_{sub.P2}$. Excessive ringing can potentially result in damage to the IC on which the power supply system **100** is provided. However, slow deactivation can result in greater switching losses, and thus less efficient operation of the power supply system **100**.

[0017] To optimize the deactivation of the power switch(es) **102**, the power supply system **100** also includes a power switch deactivation driver **104**. The power switch deactivation driver **104** can be configured to control an amplitude of the control voltage that controls the power switch(es) **102** responsive to a second state of the activation signal. The amplitude of the control voltage can be controlled based on a voltage difference between the power rail (e.g., one of the input voltage $V_{sub.IN}$ and ground) and the switching terminal to provide for a variable rate of deactivation of the power switch(es) **102**. As an example, the variable rate of deactivation can include an initially rapid deactivation of the power switch(es) **102** based on a more rapid rate of decrease of the control voltage, followed by a slower deactivation of the power switch(es) **102** based on a slower rate of decrease of the control voltage. As described in greater detail herein, the variable rate of deactivation can include three stages from full activation (e.g., saturation mode) of the power switch(es) **102** to full deactivation (e.g., cutoff mode) of the power switch(es) **102**. The variable rate of deactivation can provide for optimization of efficiency and a reduction of ringing of the current through the power switch(es) **102** during deactivation of the power switch(es) **102**.

[0018] FIG. 2 is an example diagram of a power switch deactivation driver **200**. The power switch deactivation driver **200** can correspond to the power switch deactivation driver **104** in the example of FIG. 1. Therefore, reference is to be made to the example of FIG. 1 in the following description of the example of FIG. 2. In the example of FIG. 2, the power switch deactivation driver **200** provides control of a high-side power switch, demonstrated as an NFET $N_{sub.HS}$.

[0019] The power switch deactivation driver **200** receives the input voltage $V_{sub.IN}$ at a high-voltage rail **202**, and receives an activation signal HON. The activation signal HON controls the activation and deactivation state of the power switch $N_{sub.HS}$. Responsive to an activated state of the power switch $N_{sub.HS}$, a current $I_{sub.HS}$ flows from the high-voltage rail **202** to a switching terminal **204**. The current $I_{sub.HS}$ can thus be provided through an inductor to provide the output voltage $V_{sub.OUT}$. The power switch deactivation driver **200** also receives a voltage $V_{sub.CB}$ and a voltage $V_{sub.GN1}$. The voltage $V_{sub.CB}$ can be based on a voltage $V_{sub.SW}$ of the switching

terminal **204**, such as based on being coupled to the switching terminal **204** by a capacitor (not shown). For example, the voltage $V_{sub.CB}$ can be greater than the voltage $V_{sub.SW}$ by approximately 5V. The voltage $V_{sub.GN1}$ can be provided responsive to a second state of the activation signal HON, as described in greater detail herein. As an example, the high-voltage rail **202**, the switching terminal **204**, and the voltage $V_{sub.CB}$ can be coupled to external pins of the associated IC package.

[0020] The activation signal HON is a binary signal provided through an inverter **206** to provide a signal HON'. The inverter **206** is demonstrated as biased by the high-voltage rail **202** and the switching terminal **204**. Responsive to a first state of the activation signal HON (e.g., logic-high), the power switch N.sub.HS is activated. In the example of FIG. 2, the logic-high state of the activation signal HON is inverted to provide the signal HON' at a logic-low state. The inverted activation signal HON' is provided to the gate of a P-channel FET P.sub.1. The PFET P.sub.1 has a drain that is coupled to a control node **208** and a source that is coupled to the voltage $V_{sub.CB}$. The control node **208** is coupled to the gate of the power switch N.sub.HS, such that a control voltage $V_{sub.GHS}$ controls the activation of the power switch N.sub.HS. Therefore, responsive to the logic-low state of the inverted activation signal HON', the PFET P.sub.1 is activated, thereby pulling the control voltage $V_{sub.GHS}$ to be approximately equal to the voltage $V_{sub.CB}$. Accordingly, the power switch N.sub.HS is activated to conduct the current $I_{sub.HS}$ to the switching terminal **204**.

[0021] Responsive to a second state of the activation signal HON (e.g., logic-low), the power switch N.sub.HS is deactivated at a variable rate of deactivation, as described herein. In the example of FIG. 2, the logic-low state of the activation signal HON is inverted to provide the signal HON' at a logic-high state. Therefore, responsive to the logic-high state of the inverted activation signal HON', the PFET P.sub.1 is deactivated. Approximately concurrently with the falling-edge of the activation signal HON, the voltage $V_{sub.GN1}$ is provided to activate an NFET N.sub.1 having a drain coupled to a first sense resistor R.sub.1 and a source coupled to an NFET N.sub.2 that is coupled at a source to the switching terminal **204** and is diode-connected based on having a common gate-drain coupling. Upon activation, the NFET N.sub.1 thus conducts a first sense current $I_{sub.1}$ from the input voltage $V_{sub.IN}$ through the first sense resistor R.sub.1 and through the NFETs N.sub.1 and N.sub.2 to the switching terminal **204**. Therefore, the first sense current $I_{sub.1}$ is proportional to a voltage difference between the voltage $V_{sub.IN}$ at the high-voltage rail **202** and the voltage $V_{sub.SW}$ at the switching terminal **204**.

[0022] The gate of the NFET N.sub.2 is coupled via the gate of an NFET N.sub.3, such that the NFETs N.sub.2 and N.sub.3 are arranged as a current mirror. As an example, the NFET N.sub.3 can be larger than the NFET N.sub.2 (e.g., with respect to a channel width) by a factor K. Thus, the first sense current $I_{sub.1}$ flowing through the NFET N.sub.2 is mirrored through the NFET N.sub.3 as a second sense current $I_{sub.2}$ that is proportional (e.g., by the factor K) to the first sense current $I_{sub.1}$. The second sense current $I_{sub.2}$ flows from the inverted activation signal HON', through a second sense resistor R.sub.2, and through the NFET N.sub.3 to the switching terminal **204**. The drain of the NFET N.sub.3 and the second sense resistor R.sub.2 are each coupled to a shutoff node **210** having a voltage $V_{sub.OFF}$. The gate of a shutoff switch N.sub.4 is coupled to the shutoff node **210**, with the drain of the shutoff switch N.sub.4 being coupled to the control node **208** and the source of the shutoff switch N.sub.4 being coupled to the switching terminal **204**. Therefore, the shutoff switch N.sub.4 controls the deactivation of the power switch N.sub.HS. For example, the second sense current $I_{sub.2}$ can change the amplitude of the voltage $V_{sub.OFF}$ in a variable manner responsive to the transition of the activation signal HON to a logic-low state, such that the activation of the NFET N.sub.4 in a variable rate manner provides for a variable rate of change of the control voltage $V_{sub.GHS}$. Accordingly, the power switch N.sub.HS is deactivated in a variable rate manner.

[0023] As an example, the voltage $V_{sub.OFF}$ can be expressed as follows:

$$[00001] V_{\text{OFF}} = V_{\text{CB}} - K * (V_{\text{IN}} - V_{\text{SW}}) * R_2 / R_1 \text{ Equation1}$$

Therefore, the amplitude of the voltage V.sub.OFF is based on a voltage difference VDIFF between the input voltage V.sub.IN and the voltage V.sub.SW. At larger values of the voltage difference VDIFF, the amplitude of the voltage V.sub.OFF is smaller, and at smaller values of the voltage difference VDIFF, the amplitude of the voltage V.sub.OFF is larger. As a result, the voltage V.sub.OFF can have an initial amplitude that is larger, thereby providing a rapid decrease of the control voltage V.sub.GHS and a respective rapid deactivation rate of the power switch N.sub.HS. As the voltage V.sub.SW decreases responsive to the rapid deactivation of the power switch N.sub.HS, the voltage V.sub.OFF can begin to decrease, thereby providing a slower deactivation of the power switch N.sub.HS.

[0024] FIG. 3 is an example of a timing diagram 300. The timing diagram 300 can correspond to deactivation of the power switch N.sub.HS. Therefore, reference is to be made to the example of FIG. 2 in the following description of the example of FIG. 3.

[0025] The timing diagram 300 demonstrates the input voltage V.sub.IN, the voltage V.sub.SW, the voltage V.sub.OFF, the control voltage V.sub.GHS, the current I.sub.HS, the activation signal HON, and the voltage V.sub.GN1 plotted as a function of time. At a time T.sub.1, a first stage of deactivation of the power switch N.sub.HS, demonstrated at 302, begins based on a logic-state change of the activation signal HON from logic-high to logic-low. As a result, the inverted activation signal HON' is asserted to a logic-high state. Also at the time T.sub.1, the voltage V.sub.GN1 is provided to activate the NFET N.sub.1.

[0026] Responsive to the logic-low state of the activation signal HON, and thus the logic-high state of the inverted activation signal HON', the PFET P.sub.1 is deactivated. The voltage V.sub.GN1 activates the NFET N.sub.1 to begin conducting the first sense current I.sub.1 from the input voltage V.sub.IN through the first sense resistor R.sub.1 and through the NFETs N.sub.1 and N.sub.2 to the switching terminal 204. Because the voltage difference VDIFF corresponding to the difference between the amplitude of the input voltage V.sub.IN and the voltage V.sub.SW at the switching terminal 204 is initially approximately zero, the first sense current I.sub.1 is also approximately zero.

[0027] The first sense current I.sub.1 is mirrored to generate the second sense current I.sub.2, and the second sense current I.sub.2 flowing through the shutoff node 210 generates the voltage V.sub.OFF. The second sense current I.sub.2 is also approximately zero at the time T.sub.1, which thus sets the voltage V.sub.OFF approximately equal to the voltage V.sub.CB. As demonstrated in the example of FIG. 3, and as provided by Equation 1, the voltage V.sub.OFF begins to increase during the first stage of the deactivation, resulting in a rapid decrease of the control voltage V.sub.GHS. Therefore, during the first deactivation stage 304, the power switch N.sub.HS is rapidly deactivated, providing for a higher efficiency portion of the deactivation of the power switch N.sub.HS.

[0028] At a time T.sub.2, a second stage of deactivation of the power switch N.sub.HS, demonstrated at 304, begins. During the second deactivation stage 304, as the power switch N.sub.HS deactivates, the current I.sub.HS begins to decrease. As a result, the voltage V.sub.SW at the switching terminal 204 decreases rapidly during the second deactivation stage 304. As a result, as provided by Equation 1, the voltage V.sub.OFF decreases. As a result, during the second deactivation stage 304, the control voltage V.sub.GHS levels off, thereby slowing the deactivation of the control switch N.sub.HS.

[0029] At a time T.sub.3, a third stage of deactivation of the power switch N.sub.HS, demonstrated at 306, begins. During the third deactivation stage 306, beginning at the time T.sub.3, the voltage V.sub.SW is approximately zero. Therefore, the voltage V.sub.OFF decreases to a very small amplitude as the power switch N.sub.HS deactivates. The control voltage V.sub.GHS therefore likewise decreases to fully deactivate the power switch N.sub.HS at a time T.sub.4. In the example

of FIG. 3, the input voltage $V_{\text{sub.IN}}$ is demonstrated as slightly ringing, but the ringing exhibited by the input voltage $V_{\text{sub.IN}}$ is significantly mitigated relative to a rapid deactivation of a power switch in a typical power supply system. Therefore, the example of FIG. 3 demonstrates an optimization of the deactivation of the power switch N_{sub} to provide for an initially rapid deactivation to provide for efficient deactivation to mitigate switching losses, followed by a slower deactivation to mitigate ringing.

[0030] As described above, the examples of FIGS. 2 and 3 describe a high-side power switch deactivation driver that controls deactivation of a high-side power switch. However, the power switch deactivation driver can also be arranged to control deactivation of a low-side power switch. For example, a DC-DC buck converter having a high-side power switch and a low-side power switch can include the high-side power switch deactivation driver **200** to control deactivation of the high-side power switch $N_{\text{sub.HS}}$, as well as a low-side power switch deactivation driver that controls deactivation of the low-side power switch in the same manner as described above (e.g., in a variable rate to optimize efficiency while mitigating switching losses). As another example, the power switch deactivation driver can be implemented as a low-side power switch deactivation driver for a DC-DC boost converter to control deactivation of the low-side power switch in the same manner as described above.

[0031] FIG. 4 is an example diagram of a power switch deactivation driver **400**. The power switch deactivation driver **400** can correspond to the power switch deactivation driver **104** in the example of FIG. 1. Therefore, reference is to be made to the example of FIG. 1 in the following description of the example of FIG. 4. In the example of FIG. 4, the power switch deactivation driver **400** provides control of a low-side power switch, demonstrated as an NFET NLS.

[0032] The power switch deactivation driver **400** is coupled to ground at a low-voltage rail **402**, and receives an activation signal LON. The activation signal LON controls the activation and deactivation state of the power switch NLS. Responsive to an activated state of the power switch NLS, a current $I_{\text{sub.LS}}$ flows from the low-voltage rail **402** to a switching terminal **404**. The current $I_{\text{sub.LS}}$ can thus be provided through an inductor to provide the output voltage $V_{\text{sub.OUT}}$. The power switch deactivation driver **400** also receives a voltage $V_{\text{sub.CC}}$ and a voltage $V_{\text{sub.GN5}}$. The voltage $V_{\text{sub.CC}}$ can be based on the voltage amplitude of the low-voltage rail, such as based on being coupled to the low-voltage rail **402** by a capacitor (not shown). For example, the voltage $V_{\text{sub.CC}}$ can be greater than the voltage of the low-voltage rail **402** by approximately 5V. The voltage $V_{\text{sub.GN5}}$ can be provided responsive to a second state of the activation signal LON, as described in greater detail herein. As an example, the low-voltage rail **402**, the switching terminal **404**, and the voltage $V_{\text{sub.CC}}$ can be coupled to external pins of the associated IC package.

[0033] The activation signal LON is a binary signal provided through an inverter **406** to provide a signal LON'. The inverter **406** is demonstrated as biased by the low-voltage rail **402** and the switching terminal **404**. Responsive to a first state of the activation signal LON (e.g., logic-high), the power switch NLS is activated. In the example of FIG. 4, the logic-high state of the activation signal LON is inverted to provide the signal LON' at a logic-low state. The inverted activation signal LON' is provided to the gate of a P-channel FET $P_{\text{sub.2}}$. The PFET $P_{\text{sub.2}}$ has a drain that is coupled to a control node **408** and a source that is coupled to the voltage $V_{\text{sub.CC}}$. The control node **408** is coupled to the gate of the power switch NLS, such that a control voltage $V_{\text{sub.GLS}}$ controls the activation of the power switch NLS. Therefore, responsive to the logic-low state of the inverted activation signal LON', the PFET $P_{\text{sub.2}}$ is activated, thereby pulling the control voltage $V_{\text{sub.GLS}}$ to be approximately equal to the voltage $V_{\text{sub.CC}}$. Accordingly, the power switch NLS is activated to conduct the current $I_{\text{sub.LS}}$ to the switching terminal **404**.

[0034] Responsive to a second state of the activation signal LON (e.g., logic-low), the power switch NLS is deactivated at a variable rate of deactivation, as described herein. In the example of FIG. 4, the logic-low state of the activation signal LON is inverted to provide the signal LON' at a

logic-high state. Therefore, responsive to the logic-high state of the inverted activation signal LON', the PFET P.sub.2 is deactivated. Approximately concurrently with the falling-edge of the activation signal LON, the voltage V.sub.GN5 is provided to activate an NFET N.sub.5 having a drain coupled to a first sense resistor R.sub.3 and a source coupled to an NFET N.sub.6 that is coupled at a source to the switching terminal **404** and is diode-connected based on having a common gate-drain coupling. Upon activation, the NFET N.sub.5 thus conducts a first sense current I.sub.3 from the switching terminal **404** through the first sense resistor R.sub.3 and through the NFETs N.sub.5 and N.sub.6 to the low-voltage rail **402**. Therefore, the first sense current I.sub.3 is proportional to a voltage difference between the low-voltage rail **402** and the voltage V.sub.SW at the switching terminal **404**.

[0035] The gate of the NFET No is coupled via the gate of an NFET N.sub.7, such that the NFETs N.sub.6 and N.sub.7 are arranged as a current mirror. As an example, the NFET N.sub.7 can be larger than the NFET N.sub.6 (e.g., with respect to a channel width) by a factor K. Thus, the first sense current I.sub.3 flowing through the NFET No is mirrored through the NFET N.sub.7 as a second sense current I.sub.4 that is proportional (e.g., by the factor K) to the first sense current I.sub.3. The second sense current I.sub.4 flows from the inverted activation signal LON', through a second sense resistor R.sub.4, and through the NFET N.sub.7 to the switching terminal **404**. The drain of the NFET N.sub.7 and the second sense resistor R.sub.4 are each coupled to a shutoff node **410** having a voltage V.sub.OFF2. The gate of a shutoff switch N.sub.8 is coupled to the shutoff node **410**, with the drain of the shutoff switch N.sub.8 being coupled to the control node **408** and the source of the shutoff switch Ng being coupled to the switching terminal **404**. Therefore, the shutoff switch No controls the deactivation of the power switch NLS. For example, the second sense current I.sub.4 can change the amplitude of the voltage V.sub.OFF2 in a variable manner responsive to the transition of the activation signal LON to a logic-low state, such that the activation of the NFET N.sub.8 in a variable rate manner provides for a variable rate of change of the control voltage V.sub.GLS. Accordingly, the power switch NLs is deactivated in a variable rate manner.

[0036] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action, then: (a) in a first example, device A is directly coupled to device B; or (b) in a second example, device A is indirectly coupled to device B through intervening component C if intervening component C does not substantially alter the functional relationship between device A and device B, so device B is controlled by device A via the control signal generated by device A.

[0037] Also, in this description, a device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

Furthermore, a circuit or device described herein as including certain components may instead be configured to couple to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor wafer and/or integrated circuit (IC) package) and may be configured to couple to at least some of the passive elements and/or the sources to form the described structure, either at a time of manufacture or after a time of manufacture, such as by an end user and/or a third party.

[0038] The techniques described in this disclosure may also be embodied or encoded in an article

of manufacture including a non-transitory computer-readable storage medium. Example non-transitory computer-readable storage media may include random access memory (RAM), read-only memory (ROM), programmable ROM, erasable programmable ROM, electronically erasable programmable ROM, flash memory, a solid-state drive, a hard disk, magnetic media, optical media, or any other computer readable storage devices or tangible computer readable media. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0039] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A system comprising: a switch including an input terminal coupled to a power rail, an output terminal coupled to a switching terminal, and a control terminal capable of receiving a control voltage; and a driver circuit including an output terminal coupled to the control terminal of the switch and a first input terminal capable of receiving an activation signal; wherein: the driver circuit is capable of generating the control voltage; and in response to an activation signal indicating deactivation, the switch is capable of deactivating conduction of current from the power rail to the switching terminal at a variable rate.
2. The system of claim 1, wherein: the activation signal is a logic-high signal; and an amplitude of the control voltage activates the switch to conduct the current from the power rail to the switching terminal.
3. The system of claim 1, wherein: the activation signal indicating deactivation is a logic-low signal; and an amplitude of the control voltage deactivates the switch to reduce conduction of the current from the power rail to the switching terminal.
4. The system of claim 1, wherein: the variable rate of deactivation of conduction includes a first rate of deactivation followed by a second rate of deactivation of the switch; and the first rate is greater than the second rate.
5. The system of claim 1, wherein: the switch is capable of deactivating conduction of current in three stages; a first stage includes a first deactivation rate of the switch based on an amplitude of the control voltage; a second stage includes a second deactivation rate of the switch, the second deactivation rate less than the first deactivation rate, the second deactivation rate based on the amplitude of the control voltage; and a third stage includes a third deactivation rate that is complete deactivation of the switch responsive to a largest voltage difference between the power rail and the switching terminal.
6. The system of claim 1, wherein: the switch is a high-side power switch of a DC-DC buck power supply system.
7. The system of claim 1, wherein: the driver circuit includes an inverter with an input and an output; and the first input terminal of the driver circuit is coupled to the input of the inverter.
8. The system of claim 7, wherein: the driver circuit includes a first resistor including a first terminal and a second terminal; the driver circuit includes a second resistor including a first terminal and a second terminal; and the first terminal of the first resistor is coupled to the output of the inverter and the first terminal of the second resistor.
9. The system of claim 8, wherein: the driver circuit includes a first transistor including a first terminal, a second terminal and a control terminal; and the first terminal of the first transistor is coupled to the second terminal of the first resistor.
10. The system of claim 9, wherein: the driver circuit includes a second transistor with a first terminal coupled to the second terminal of the first transistor, a second terminal, and a control terminal; and the driver circuit includes a third transistor with a control terminal coupled to the

- control terminal of the second transistor, a first terminal coupled to the second terminal of the second resistor, and a second terminal coupled to the second terminal of the second transistor.
- 11.** The system of claim 10, wherein: the driver circuit includes a fourth transistor with a control terminal coupled to the first terminal of the second resistor, a first terminal, and a second terminal coupled to the inverter.
- 12.** The system of claim 11, wherein: the driver circuit includes a fifth transistor with a control terminal coupled to the second terminal of the second resistor, a first terminal coupled to the first terminal of the fourth transistor, and a second terminal coupled to the second terminal of the third transistor.
- 13.** The system of claim 12, wherein: the output of the driver circuit is coupled to the first terminal of the fourth transistor and the first terminal of the fifth transistor.
- 14.** A circuit comprising: a sense resistor coupled to a power rail capable of conducting a sense current having an amplitude based on a voltage difference between the power rail and a switching terminal responsive to a state of an activation signal; and a shutoff switch capable of controlling an amplitude of a control voltage capable of controlling an activation of a power switch, the shutoff switch being controlled by the amplitude of the sense current to provide for a variable rate of deactivation of the power switch.
- 15.** The circuit of claim 14, wherein: the variable rate of deactivation of the power switch comprises a first rate of deactivation of the power switch followed by a second rate of deactivation of the power switch.
- 16.** The circuit of claim 15, wherein: the first rate of deactivation is greater than the second rate of deactivation of the power switch.
- 17.** The circuit of claim 14, wherein: the sense resistor is a first sense resistor capable of conducting a first sense current responsive to the state of the activation signal; and the power switch includes: a second sense resistor capable of conducting a second sense current responsive to the state of the activation signal; and a current mirror capable of mirroring the first sense current to generate the second sense current through the second sense resistor as proportional to the first sense current, wherein the shutoff switch comprises an input terminal coupled to a shutoff node through which the second sense current flows, such that the shutoff switch is controlled by the amplitude of the second sense current.
- 18.** The circuit of claim 17, wherein: the second sense current flows from the activation signal to the power rail or the switching terminal.
- 19.** The circuit of claim 14, wherein: the deactivation of the power switch occurs in three stages responsive to the activation signal indicating a deactivation state; a first stage includes a first rate of deactivation of the power switch based on the amplitude of the control voltage; a second stage includes a second rate of deactivation of the power switch based on the amplitude of the control voltage; and a third stage includes a third rate of deactivation of the power switch responsive to a largest voltage difference between the power rail and the switching terminal.
- 20.** The power switch of claim 19, wherein: the first rate of deactivation is greater than the second rate of deactivation; and the third rate of deactivation is complete deactivation of the power switch.
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